

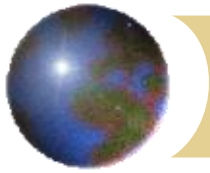
ThaiGrid: Current Status

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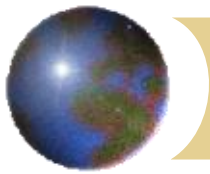
Motivation

- ✚ Sharing of High Performance Computing Resources
- ✚ Explore Grid Technology and Software Tools
 - ▣ Grid Software Tools, Portal, Scheduler
- ✚ Explore Grid Applications
 - ▣ Bioinformatics
 - ▣ Computational Fluid Dynamics
 - ▣ Collaborative Computing



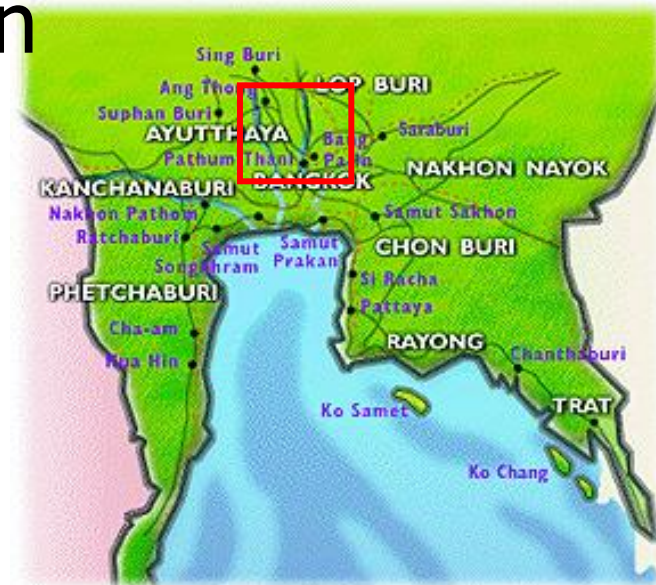
Outline

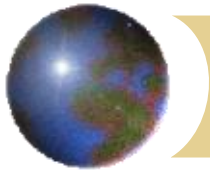
- ✚ Computational Grid
 - ✚ Deployment of Grid technologies
 - ✚ Make the system operational
- ✚ Collaborative Grid
 - ✚ Provide the Access Grid to be usable and accessible



ThaiGrid Requirements

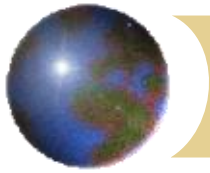
- ✚ Research and development in Grid Computing
- ✚ Single Sign on Capability
- ✚ Batch Serial Job Submission
- ✚ Parallel Job Submission
- ✚ Bulk Data Transfers





Accomplished Works

- ✚ Define the basic architecture of ThaiGrid
- ✚ Multi-site MPI test
- ✚ Setup infrastructure to issue Thaigrid certificate
 - ✚ Thank to the assistant of Globus Project (Ian Foster)



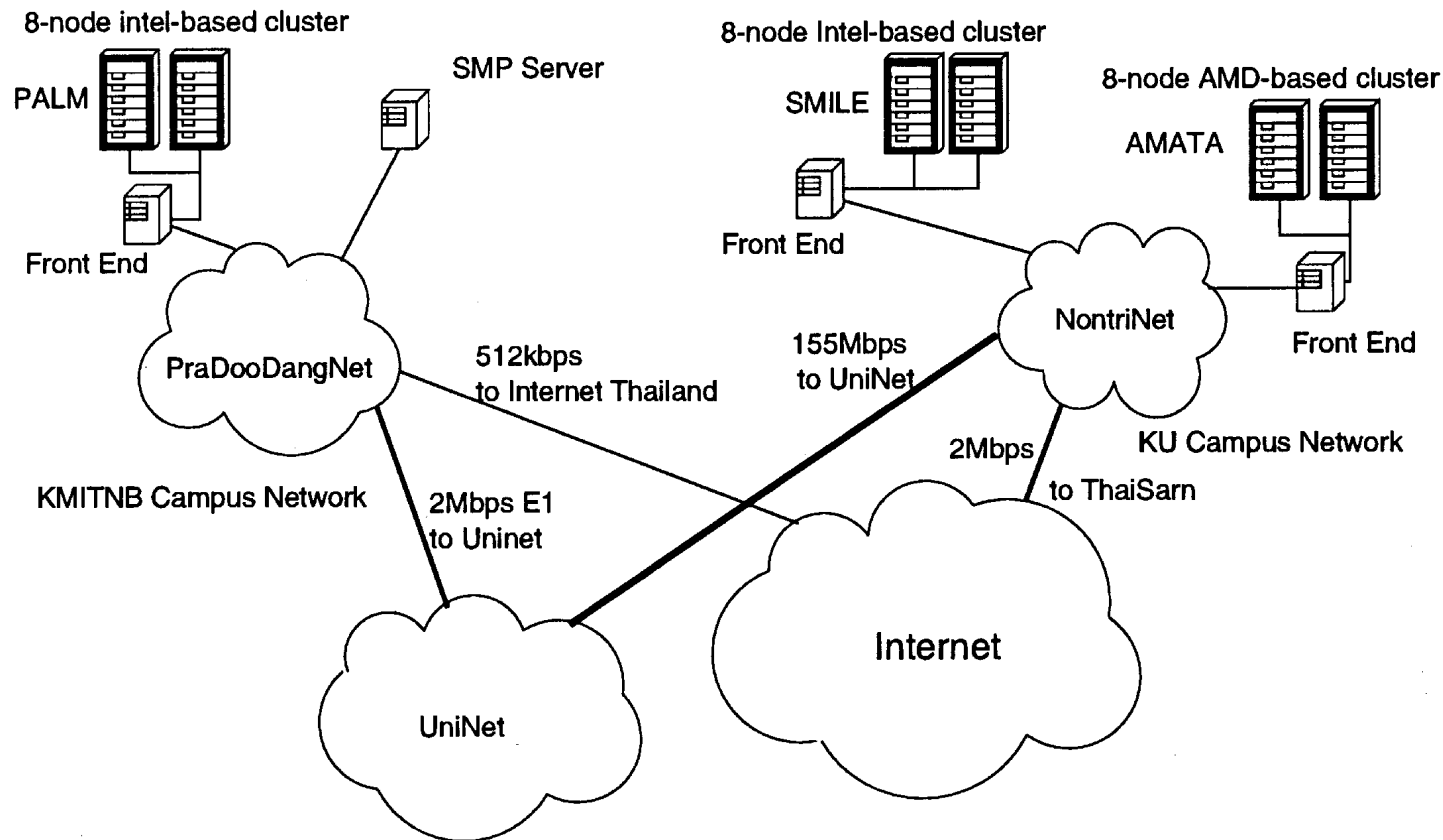
Accomplished Works

- ✚ Deploy Globus 1.1.4
- ✚ Over 106 nodes testing
- ✚ Startup Simple Grid Portal at KU.
- ✚ Operational Access Grid

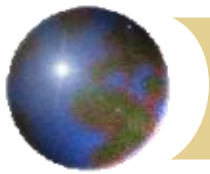


ThaiGrid Initial Testbed

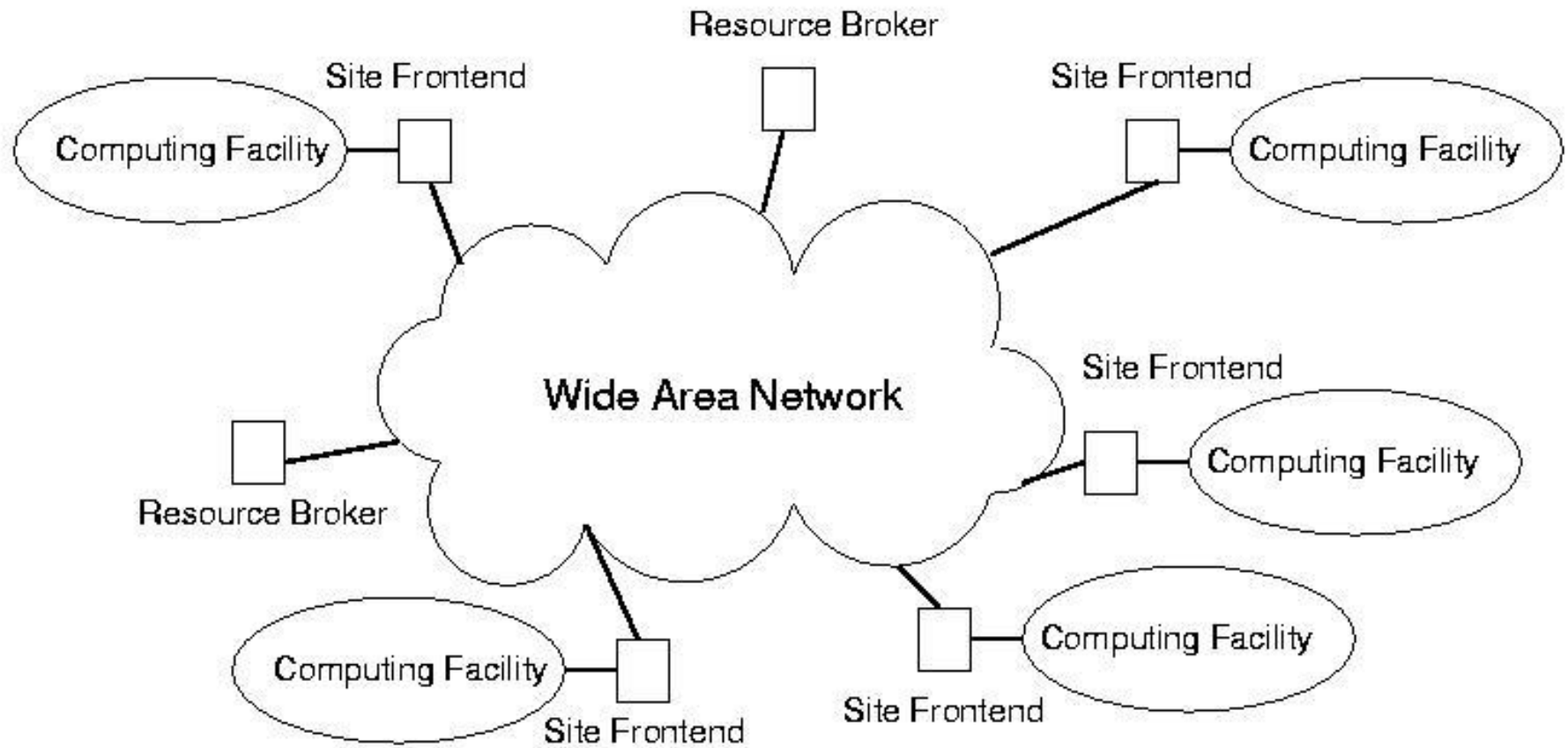
(Jan 2000)

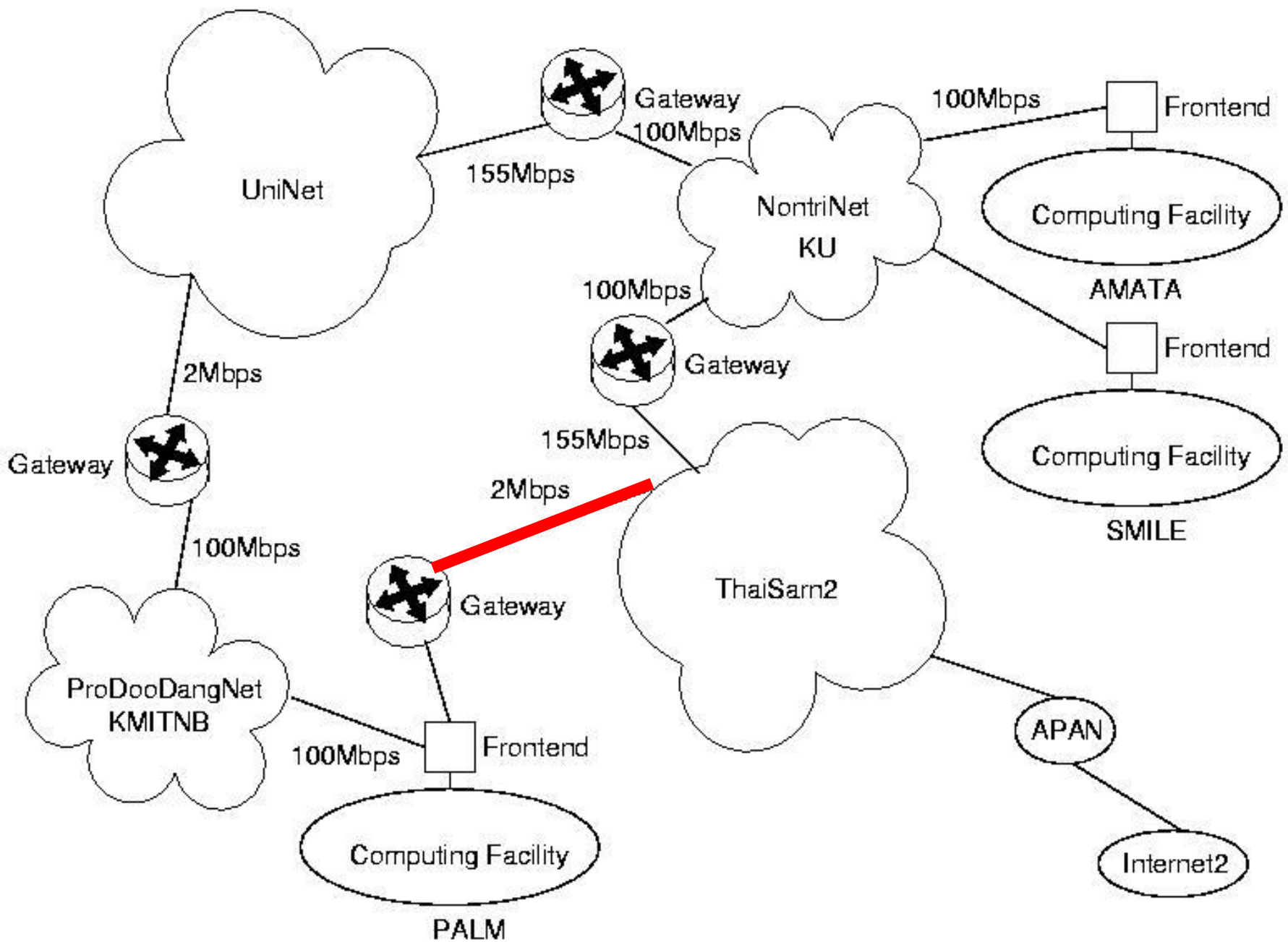


The ThaiGrid testbed topology.

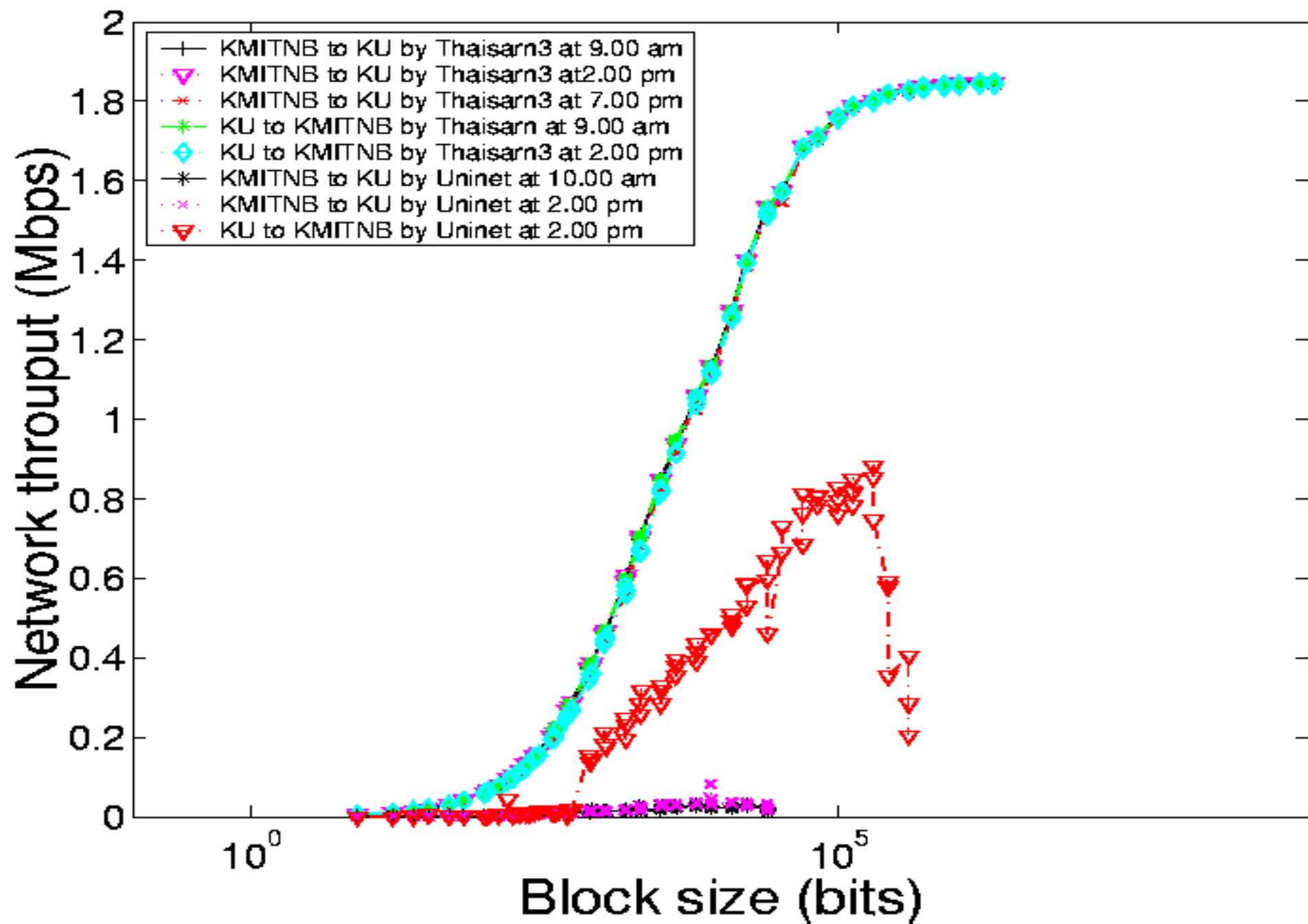


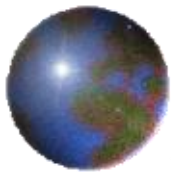
ThaiGrid Architecture





Test KMITNB and KU link





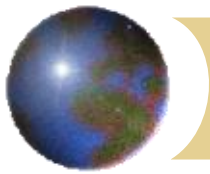
Next Phase of ThaiGrid





Middleware for ThaiGrid

- ✚ Globus version 1.1.4
- ✚ SSL
- ✚ LDAP
- ✚ MPICH with G2 device enable
- ✚ Condor Suite



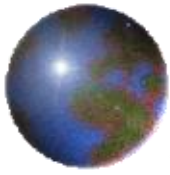
Current Computing Systems on ThaiGrid

- ✚ Pirun Beowulf Cluster
 - ▣ 75 nodes PIII500 Mhz and Xeon (72 on ThaiGrid)
- ✚ AMATA
 - ▣ 16 Athlon 1 GHz (8 myrinet port and Fast Ethernet)
- ✚ SMILE
 - ▣ 10 or PPRO200, Pentium II 350, PII450 (6 on Thaigrid)
- ✚ PSI
 - ▣ 3 Pentium II 350 Mhz
- ✚ Prgservice
 - ▣ 1 of PII350
- ✚ PALM
 - ▣ 8 nodes PentiumII 400 MHz.
- ✚ **Total of 106 nodes is now usable on Thaigrid**



On Going Works

- ✚ Operate the batch jobs submission based on Condor
- ✚ Operate a simple grid portal
- ✚ Estimate the performability of the ThaiGrid
- ✚ CFD: more by Prof. Putchong



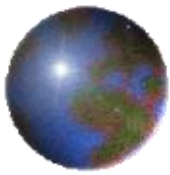
Collaborative Grid

APAN 2002 Phuket Thailand Jan
2002



TAG: Thai Access Grid

- ✚ Utilizing COTs
- ✚ Based on Multicast Technology
 - ▣ mBone
 - ▣ Multicast by Hardware Router
- ✚ vic
- ✚ OMEGA project
 - ▣ Open Multi-Environment Grid Adapter

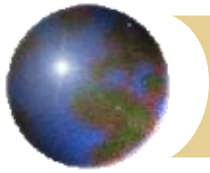


- ✚ KU
- ✚ KMITNB
- ✚ Sydney VisLab
- ✚ NUS



APAN 2002 Phuket Thailand Jan
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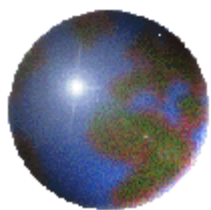
Future Works

- ❖ Encourage more participation to ThaiGrid Project
- ❖ Utilizing **Faster** ThaiSarn2 and Uninet as a Grid Backbone
- ❖ Implementing full set of services for high performance computing
- ❖ Collaborate with International Grid Computing Effort and Link with other grid



Research Issues

- ✚ SCE--Middleware
- ✚ Access Grid Bridge
- ✚ Low Cost Access Grid Point
- ✚ Utilizing Grid Technology under limited bandwidth
- ✚ Performability Study



More Information

URL: <http://www.thaigrid.net>